Lunar tides across an interrupted coastal–inland optical-clock link

[Authors, affiliations and correspondence: author confirmation required.]

Optical clocks probe gravitational potential through their relative rates.[1,2] On a coastal–inland baseline, the predicted tidal signal combines the solid-Earth response with substantial ocean loading. Here we estimate the lunar response in an interrupted strontium–ytterbium comparison over the 1,350-km Shanghai–Wuhan fibre route.[3] Six operating periods in 2026 provide 20.1 nominal accumulated sample days. A common M2+N2 scale is 0.973 ± 0.121, with the quoted working error equal to the largest of least-squares, delete-day and delete-period standard errors. The scale changes little when constant levels are assigned by period, group or segment. Separating the constituents identifies M2 as the better-constrained response; N2 is not individually resolved. Period-specific solar-day terms shift the common scale to 0.761 ± 0.205; conditional six-period scalar tests give $P = 0.03125$ at both zero and unity, without adjustment for exploratory model choice. A whole-template amplitude learned from other periods reduces held-out residual sums of squares after local nuisance fitting. The comparison complements contemporaneous European network tracking[4] with a load-sensitive test across interruptions. The fitted lunar response is primarily constrained by M2; unresolved time-dependent clock and transfer effects delimit its physical attribution.

The gravitational redshift links the rate of an atomic clock to its gravitational environment. Optical clocks have enabled laboratory tests of this relation, centimetre-scale geodesy and resolution of a potential gradient across a millimetre atomic sample.[5,6,7] Transportable clocks and coherent optical networks extend these measurements beyond one laboratory.[8,9,10] Time-variable potential adds a complementary observable: astronomical forcing, deformation of the Earth and redistribution of ocean water change the rates of clocks at geographically separated sites.[11,12,13]

Coastal and inland clocks sample different combinations of astronomical forcing and the Earth's response. Ocean mass redistribution adds a direct gravitational contribution and deforms the land supporting the clocks. A coastal–inland comparison therefore provides a physical test in which loading accounts for a substantial part of the prediction, rather than a small correction to an otherwise complete solid-Earth signal.[14,15,16,17] The challenge is to retain this time-dependent information when the absolute frequency level changes between operating periods.

We study this problem on a coastal–inland baseline connecting a Sr clock in Shanghai to a Yb clock in Wuhan. The related frequency-ratio experiment and the evaluations of the clock systems provide the apparatus context.[3,18,19] Long-distance coherent dissemination has been demonstrated in several network configurations; published performance on other links is not assigned to this record.[20,21] According

to the experimental team, the observations were acquired independently of the contemporaneous European network measurements of Stahl and colleagues.[4] This analysis was developed with knowledge of their work. Their network study reports within-day tracking on two geographical configurations, using a constant tidal amplitude and an evolving frequency baseline in its Kalman estimator. Both studies include ocean loading and estimate a constant tidal response. On our interrupted coastal–inland record, we examine constituent sensitivity and amplitude transfer; 1,350 km denotes the fibre-route length. The data extend overlapping summer observations with other 2026 operating periods; they are not a second independent experiment on the same link.

A phase-diverse record with measured support

Six operating periods between 7 January and 26 August 2026 contain 1,737,073 retained nominal one-second samples in 49 effective intervals. Their 2,905 non-overlapping 600-s averages span 231 calendar days and accumulate 20.1 nominal sample days (Fig. 1b; Extended Data Table 1). Source membership is preserved, and each theoretical average uses the same members as its observation. These exposures describe repeated measurements of one clock comparison, rather than uninterrupted operation or six independent experiments.

The 2025 commissioning observations used different frequency conversion and processing conventions, with unresolved level corrections. They are treated separately within their September and October stages and do not enter the common 2026 response. This scope follows the experimental history rather than the size of a fitted amplitude.

The prediction combines solid-Earth and ocean-loading contributions under a documented IERS-based construction and FES2022 loading inputs.[14,15,16,17] On the retained 2026 sampling, the demeaned root-mean-square contributions are 3.245×10^{-18} and 1.507×10^{-18}, with a combined value of 4.410×10^{-18}. The loading-to-total RMS ratio is 34.2% (Fig. 1a); it is not a variance fraction because the contributions are correlated. The main response estimate tests this combined prediction. The size of the loading term motivates a geographically complementary test; it is a predicted contribution on the retained sampling, rather than a measured loading fraction. An exploratory fit that allows separate solid-Earth and loading scales gives a loading response of 0.937 ± 0.529 (Methods). The record therefore tests a load-sensitive prediction without yet resolving ocean loading separately or measuring a Love number.

The approximately 12.42-h M2 period advances through the solar day. Its changing phase relative to an exact 12-h oscillation supplies information that is absent from a single short observation window. Figure 1c shows the actual local-hour and astronomical-phase coverage. A large calendar span alone does not guarantee separation: allowing a disturbance amplitude and phase to reset frequently removes part of the information supplied by this evolution.

A static physical response across operating periods

Our reference model estimates one constant tidal scale and one constant level in each recorded operating period:

$$y_i = A_T T_i + c_{p(i)} + \epsilon_i.$$

Here T_i is the fixed, member-matched tidal prediction. The six c_p account for unknown constant levels; they are not independently measured calibration states. All parameters are fitted simultaneously to the unchanged observations. There is no preliminary alignment of segment centres to the tide, no time-dependent fitted tidal amplitude and no fit-spliced stability calculation.

The resulting scale is $A_T = 0.913 \pm 0.105$. The quoted working error is the largest of residual-scaled least-squares, delete-BJT-day and delete-period errors, reported separately in Extended Data Table 2. A conditional six-period sign test gives $P = 0.03125$ against zero and $P = 0.5625$ against unit response. The first value reaches the resolution limit of the 64-pattern, two-sided enumeration; it is not a Gaussian detection significance or an experimentally calibrated tail probability.

Replacing period levels by operating-group or individual-segment levels gives whole-template scales of 0.885 ± 0.109 and 0.832 ± 0.120. These changes test how the response depends on level freedom. They do not certify that missing corrections are constant within a period. The period reference was specified after inspection of earlier analyses, rather than prospectively preregistered.

The response also transfers between periods. Training the scale on five periods and fitting only the nuisance level in the held-out period improves residual sums of squares in five of six periods, with the sixth changing by less than 0.02% (Fig. 3b). The pooled reduction is 3.53%, and the six training scales span 0.864–1.002. These are conditional amplitude-transfer tests: the held-out level is fitted locally and the training sets overlap. A complementary early/late split transfers amplitudes between two non-overlapping three-period groups. All four declared designs reduce test-set RSS in both directions, with reductions of 1.10% and 6.62% for the whole-template reference (Extended Data Table 3). Other test-period coefficients are estimated locally; the two directions reuse the same record. Positive transfer gains show predictive value relative to the specified zero-amplitude comparator; they do not establish equal tidal amplitudes between periods.

Lunar amplitude and phase

We split the deterministic prediction into M2+N2, S2+K2 and a diurnal group, preserving the predicted constituent proportions within each group. The remaining theory is retained once as a fixed remainder. With the same six period offsets, the lunar coefficient is

$$A_L = 0.973 \pm 0.121.$$

The near-12-h and diurnal coefficients are 1.005 ± 0.271 and 0.462 ± 0.415. K2 is lunisolar, so the near-12-h coefficient is not a pure solar-gravity measurement. Changing the offset resolution gives lunar scales of 0.945 ± 0.123 for operating groups and 0.952 ± 0.137 for effective segments (Fig. 2a).

An exploratory separation of M2 and N2 clarifies what supports the common lunar response. The period-offset estimates are $A_{M2} = 0.997 \pm 0.123$ and $A_{N2} = 0.541 \pm 0.648$ (Extended Data Table 4). The stronger M2 component agrees with the expected response; N2 is not separately resolved. Projected subtraction and joint estimation agree when uncertainty in the fitted strong component is retained (Methods).

A two-coefficient lunar fit tests phase without shifting the observation times. The in-phase and quadrature estimates are $(0.972, -0.070)$ (Fig. 2b). The unit, in-phase prediction is consistent with the working-model regions. The corresponding two-parameter six-period test gives $P = 0.03125$ against zero and $P = 0.90625$ at the unit, in-phase point. The phase plot and scalar estimate use the same observations and are not independent confirmations. An additional M2-only phase test, with N2 fitted separately, gives in-phase/quadrature coefficients $(0.998, -0.068)$ without daily terms and $(0.821, -0.009)$ with period-specific daily terms. Both are compatible with the unit, in-phase prediction under their conditional joint tests. The latter test does not exclude zero at 5%; agreement with a predicted point and exclusion of zero are distinct questions.

Solar-day sensitivity is assessed by adding exact 24-h and 12-h sine/cosine pairs to the joint model. A single pair set shared across the record gives $A_L = 1.023 \pm 0.123$; independent pairs in each period give 0.761 ± 0.205. The latter design retains 38.6% of the nuisance-projected lunar-template squared norm in the period-offset reference. More frequent ten-day and five-day resets give 0.891 ± 0.282 and 0.893 ± 0.430. Their scalar period-test probabilities are 0.0625 and 0.09375, whereas the period-daily model reaches 0.03125. Compatibility with unit response is also model-dependent: the period-specific daily scalar model gives $P = 0.03125$ at unity, whereas the corresponding joint phase test gives $P = 0.125$. Thus zero rejection does not by itself establish unit-response agreement. These probabilities need not vary monotonically with parameter count, because the restricted residuals and studentizing covariance also change.

The physical interpretation is specific. An unconstrained exact 12-h disturbance cannot be separated from a pure S2 response using the same frequency series alone. M2 and N2 have different frequencies and retain measurable template directions, although short-window daily freedom removes information. The free daily coefficients therefore describe a specified sensitivity class, not measured environmental states. These sensitivity models quantify dependence on assumed disturbance freedom; they neither identify a contamination mechanism nor bound unmodelled variation in the reference fit. Independent

bounds on clocks and unmonitored transfer branches are needed to turn these tests into a complete systematic-error budget.

Physical scope and complementary measurements

The predesignated B2 round-trip monitor tests its recorded transfer channel. On 1,519 fully matched summer bins from P6–P8, adding B2 to the period-offset model changes the whole-template and lunar scales by 0.0057 and 0.0172, with paired delete-day standard errors of 0.0072 and 0.0094. All three archived monitor thresholds and all four declared designs are retained (Extended Data Table 5). These same-support association checks do not bound the six-period result or unmonitored clock effects and laboratory branches. No same-site null comparison is available.

The reference-model response is consistent with the expected tidal scale and remains similar across the recorded group and segment level partitions. This agreement is conditional on the applied correction coordinate: offset comparisons do not bound unrecorded within-period variation. Its substantial predicted contribution on the Shanghai–Wuhan baseline motivates testing the combined response and asking whether a separate loading scale is identifiable. The present record constrains the combined response more strongly than that separate scale.

The common response is not a measurement of seasonal potential or an absolute connection between operating periods. Slow signals require calibrated levels that the short-period offsets intentionally discard. Similarly, agreement with a combined template does not independently verify every loading component, station coordinate or frequency-dependent response. The associated frequency-ratio preprint reported a different whole-template estimator on overlapping summer observations; its $0.87(15)$ and reported 5.7σ are not significance statements for the present lunar fit.[3] Relative to that overlapping summer analysis, this study adds other 2026 operating periods, distinguishes the dominant M2 contribution from unresolved N2, and tests the transfer of tidal amplitude across interruptions. The record constrains a predominantly M2 response and demonstrates conditional amplitude transfer across interruptions; N2 and a separate loading response remain unresolved. This is a constituent-level test of a load-sensitive prediction, with the model dependence quantified on the same observations.

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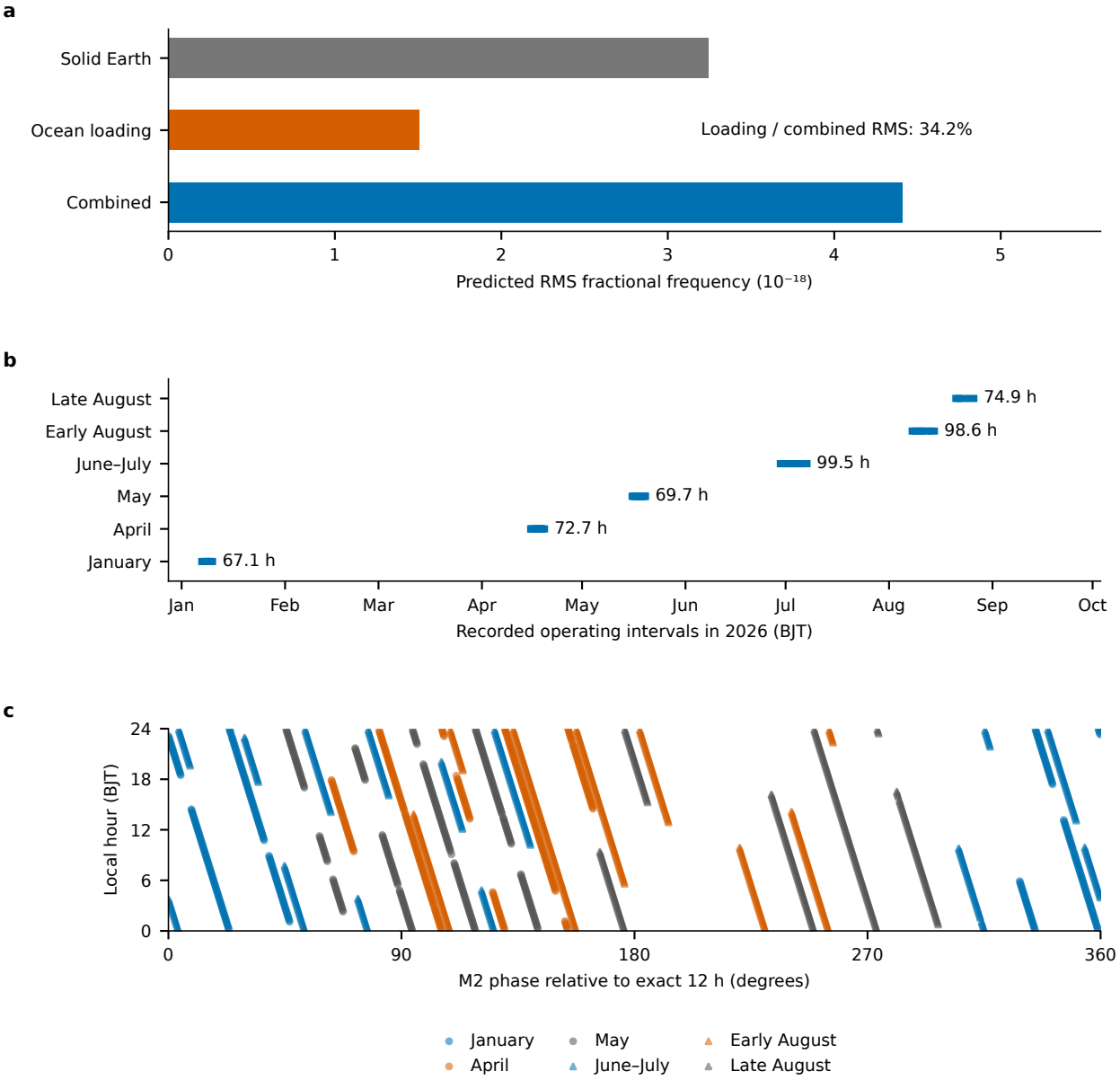


Fig. 1 | Physical prediction and sampling on the coastal–inland comparison. **a,** Globally demeaned RMS of the deterministic components on the retained sampling. Correlated components have non-additive RMS values; the loading/combined ratio is a prediction, not a measured loading fraction. **b,** Effective-interval spans represented by their first and last bin-mean times, annotated by retained nominal member-hours, not continuous uptime. **c,** Local hour versus M2 phase relative to an exact 12-h oscillation, from the averaged basis columns. All 2,905 averages are shown. These panels describe prediction and sampling before measurement fitting; phase coverage alone does not establish environmental separation.

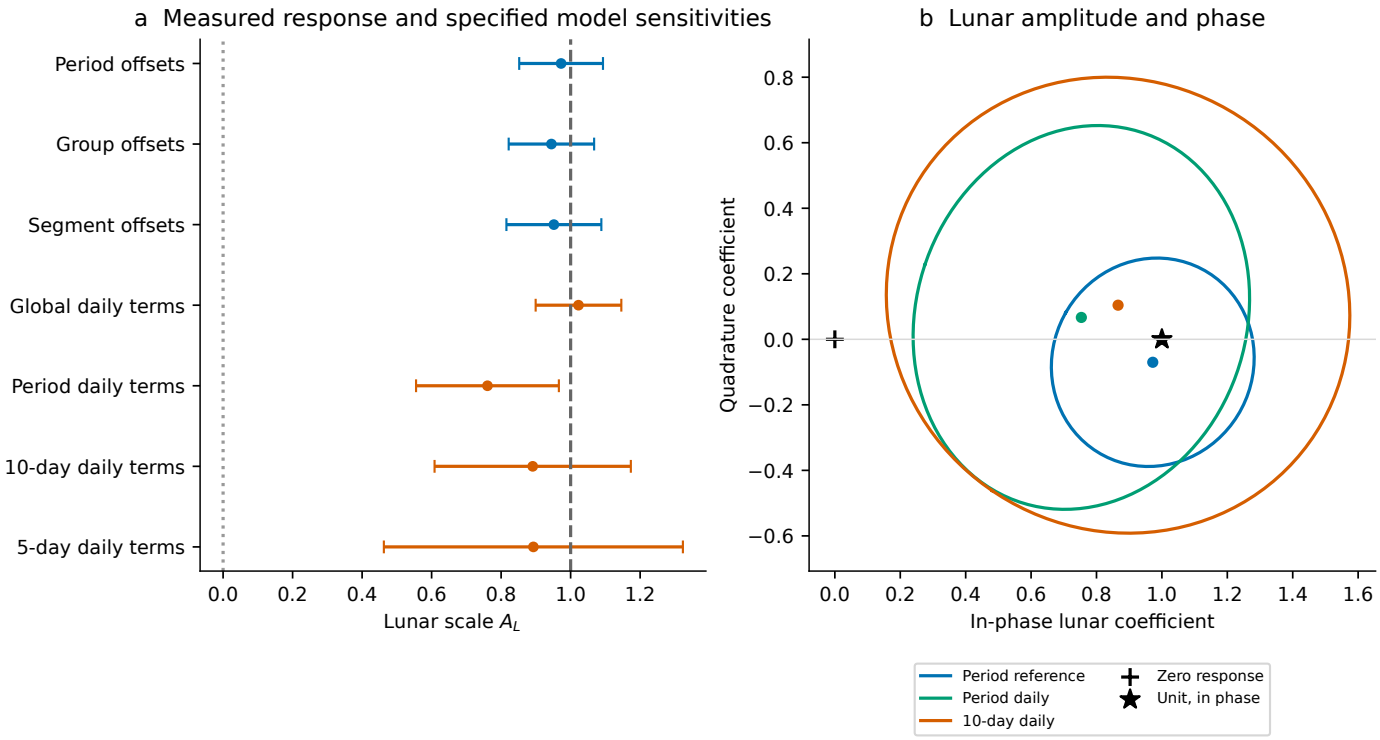


Fig. 2 | Lunar response under specified level and disturbance models. a, M2+N2 scales on identical observations. Bars are one working standard-error summary, max(LS, delete-day, delete-period), not calibrated 95% confidence intervals. The first three rows change only level resolution; later rows introduce specified daily freedom. **b,** In-phase and quadrature lunar coefficients. Curves are nominal 95% delete-day F regions with 34 denominator degrees of freedom. The unit prediction lies within all displayed regions; zero lies outside these day-based contours. This does not imply identical period-level inference: the corresponding joint sign-test probabilities at zero are 0.03125, 0.03125 and 0.25 for the period reference, period-daily and ten-day-daily models. All use the same observations, with no time shifts.

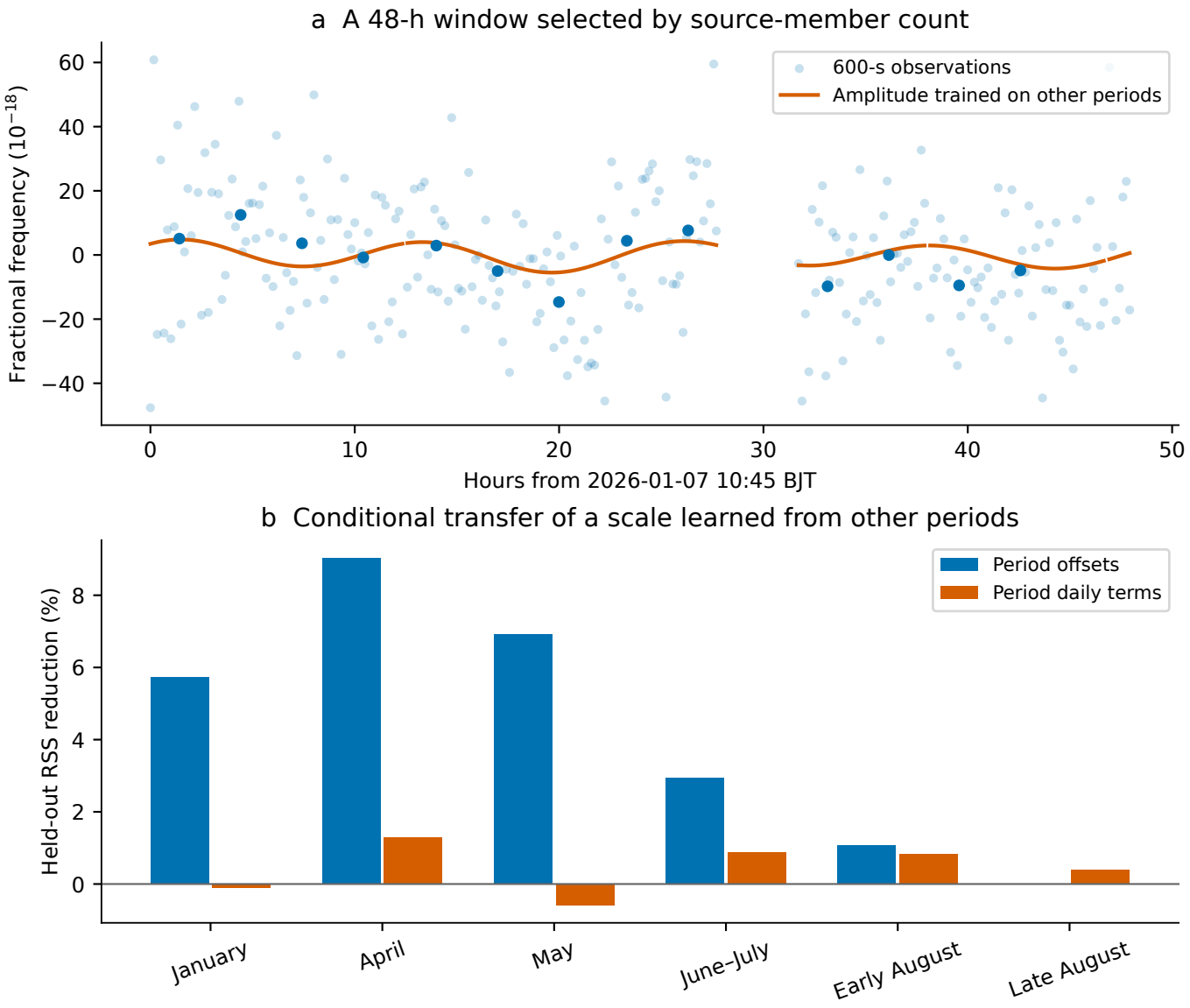


Fig. 3 | A learned scale tested on held-out periods. a, The 48-h window with the largest retained source count: 158,072 members in 264 averages. Pale points show 600-s observations; dark points average 18 consecutive bins within uninterrupted effective intervals. The orange curve uses the template scale learned from the other five periods. A single level fitted across the held-out January period is removed for display; no segment centres are independently shifted. **b,** Held-out RSS reductions relative to the same locally fitted nuisance model with zero tidal scale. Negative bars are retained. The six folds share training data and supply conditional amplitude-transfer diagnostics, not independent detections.

Methods

Physical quantities, nuisance roles and model review

The observable is $y = (R/R_{\mathrm{ref}} - 1)10^{18}$, with $R = f_{\mathrm{Yb,WUH}}/f_{\mathrm{Sr,SHA}}$ and $R_{\mathrm{ref}} = 1.2075070393433377166$. Positive y means that Wuhan Yb is faster relative to Shanghai Sr. For positive Newtonian potential U, the fractional gravitational contribution is $-[\delta U_{\mathrm{WUH}} - \delta U_{\mathrm{SHA}}]/c^2$; templates are converted to the same 10^{-18} working units as y before regression. A template scale tests a specified prediction; it is not identified with an independently measured Love number or with a stochastic trajectory.

Tidal scales and constant level offsets are estimated jointly. Their physical and statistical roles are summarized in Extended Data Table 6. Offset indicators represent unknown constant levels within recorded periods, groups or segments, conditional on within-group constancy. Daily sinusoids define empirical sensitivity models and are not interpreted as measured environmental states. A state-space estimator can also be appropriate when its transition, observation and noise assumptions are justified.[22]

The reference design was specified after earlier model comparisons. All twenty declared reference, component and sensitivity designs are retained; no model was selected by proximity to unity or smallest probability. The reference uses the six recorded 2026 operating periods. Group, segment and fixed-calendar partitions test its dependence on level resolution. A statistical nuisance parameter is not thereby an observed physical process.

Frequency conversion and frozen observations

The chain uses $N_{1397} = 858456$, $N_{1550} = 773598$, $N_{1156} = 1295739$ and $N_{1550,\mathrm{WH}} = 966996$. The recorded 2025 relationship gives

$$\alpha = \frac{10.93629485}{25N_{1397}}, \qquad f_{1397} = \frac{f_{\mathrm{Sr}}}{2(1-\alpha)}, \qquad f_{1550,\mathrm{WH}} = \left(N_{1550} + \frac{8}{25}\right)\frac{f_{1397}}{N_{1397}}.$$

The eight reference units include seven on transmission and one at Wuhan. For recorded counter value q in nominal hertz realized by the Wuhan reference,

$$R_{\mathrm{lock}} = \frac{N_{1156}}{N_{1397}}\frac{N_{1550} + 8/25}{N_{1550,\mathrm{WH}} + 26/20 + dq/(20\times 10^7)}\frac{1}{1-\alpha}.$$

The inherited 2026 configuration uses $d = 1$ and $\alpha = 0$, whereas the separately analysed 2025 records use $d = 2$ and the stated α. For signed Sr and Yb corrections a, b, the reconstruction uses $R = R_{\mathrm{lock}}(1+a)/(1+b)(1+\Delta G)$ with inherited $\Delta G = -3.116\times 10^{-15}$. Decimal anchors precede conversion to floating point; full ratios are not subtracted in float64.

The corrected experimental intervals and existing theory-independent screen are frozen. Timestamp

selection uses the recorded start inclusively and the recorded end plus one second exclusively. Repeated source identities at shared interval endpoints are retained once; repeated timestamp labels from distinct rows remain. Finite beat offsets within 10^6 Hz of the nominal anchor define each interval's median; rows must lie strictly within 10 Hz of that median. The parent selection retains 1,966,908 rows at this stage.

Screening then forms 300-s blocks from the first timestamp surviving that gross screen in each effective interval. Block means are compared with a 13-block running median, with reflected boundaries and empty blocks treated as missing. The scale is the median absolute deviation of finite block residuals about their median, divided by 0.6744897501960817. Blocks whose absolute residual exceeds six scales and their immediately adjacent blocks are removed. No tidal residual enters this rule. For final averaging, the separate origin is the floored earliest selected timestamp in the interval before quality screening. Half-open, non-overlapping 600-s bins require at least 300 surviving members, including at interval ends; shorter and empty bins are omitted. Observations and interpolated theory are averaged over identical retained source rows.

The 2026 subset contains 1,737,073 members in 2,905 averages. Observations and theory have identical source membership. A separate raw-counter readback previously checked these means, and its identities and SHA256 records accompany this revision. The new fit reads that independently reconstructed response, not any fit-adjusted series. Raw-source readback verifies implementation of the specified conversion and aggregation, conditional on the inherited frequency-chain constants and correction ledger. It covered the selected 2026 members in 34 contributing files, not an independent certification of gate duration, synchronization, selection or correction values. BJT labels are numerical time coordinates; astronomical evaluation applies the documented UTC conversion. Exposure is counter-row count times the nominal one-second gate, including repeated timestamp labels. The 600-s averages contain 300–602 members; member count is neither unique physical seconds nor statistically independent sample size.

The applied 2026 correction values and missing entries are listed in Extended Data Table 9. The ledger uses a signed optical-offset convention: the Sr contribution enters the ratio through $1 + a$, and the Yb contribution through $1 + b$ in the denominator. The table records what was applied computationally; it does not establish whether each shift was already compensated online, its calibrated optical node, or its constancy within an operating period. Unknown clock corrections remain unknown rather than measured zero. Early Sr AC entries and the May Yb correction lack complete calibration. A 37.2-mHz feedback entry at the assumed 698-nm optical node corresponds to 86.6119×10^{-18} for the affected historical ledger entries; their mapping and optical feedback node are not newly certified here. A constant contribution is absorbed by an intercept, but time dependence is not. The historical Yb correction near -1.1×10^{-17} is not added again without resolving overlap with the updated ledger. No correction, mask or observation was changed to improve this revision's agreement. Synthetic period-constant shifts of $(-11, 86.6119, 11, -25, 50, -50) \times 10^{-18}$ change the whole-template and grouped lunar scales

by less than 10^{-14} in this design. This checks additive intercept invariance, not the physical accuracy or inclusion of those corrections. In contrast, adding 0.1 times the target template changes its scale by 0.1: a target-aligned unmonitored variation is inseparable from response in a single channel. These synthetic checks do not bound actual time-dependent or multiplicative errors.

Deterministic tidal synthesis and its limits

The frozen theoretical series combines an IERS-based solid-Earth calculation and FES2022 loading inputs.[14,15,23] Pysolid radial displacements are converted to fractional frequency through

$$s(t) = -\frac{1 + k_2 - h_2}{h_2} \frac{g[u_{\mathrm{WUH}}(t) - u_{\mathrm{SHA}}(t)]}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\ \mathrm{m\,s^{-2}}$. This is a displacement-to-potential approximation, not a complete degree- and frequency-dependent potential calculation. The archived loading construction includes direct load potential, induced potential and displacement through load-Love responses.[16,17] The eighteen-constituent loading input does not supply a complete long-period load model; atmospheric loading has not been independently modelled or bounded. The documented station coordinates are theory-input coordinates rather than newly certified clock-transition locations.

The complete 360,001-point, 90-s theory grid, from 1 September 2025 to 11 September 2026 in the recorded BJT coordinate, is decomposed by equally weighted least squares into quadratures of Q1, O1, P1, K1, N2, M2, S2, K2, Mf, Mm, Msf, Sa and Ssa plus a constant, retaining the remainder. Astronomy uses the recorded FES convention and TT−UTC=69.184 s; Beijing time is converted to UTC once.[24] Decomposition is applied to the deterministic theory, not learned from clock residuals. Linear interpolation on that grid precedes source-member averaging, without extrapolation. The full model scales the total series. Component models jointly fit M2+N2, S2+K2 and K1+O1+P1+Q1, with the remaining prediction subtracted exactly once.

Earlier DE430 geometric-potential and finite-basis frequency-response comparisons supply named sensitivity checks, not certification of all template errors.[25] Their published numerical shifts belong to their original designs and are not copied as uncertainty bounds for the new period reference. The reported template scale remains conditional on the frozen physical construction. The frozen loading phase and sign convention is retained without amplitude or phase adjustment against the observations.

Loading sensitivity and the historical estimator

For the same 2,905 member-matched observations, an exploratory physical decomposition uses $y_i = A_{\mathrm{E}}E_i + A_{\mathrm{O}}O_i + c_{p(i)} + \epsilon_i$, where E is the solid-Earth template and O the loading template. Their sum agrees with the frozen total to 1.3×10^{-32} in fractional frequency. Globally demeaned template

correlation is 0.680. After removing period constants, the correlation is 0.686 and 52.9% of the already period-projected loading-template energy remains after removing the solid-Earth direction. Neither quantity is an observed loading fraction or a measured improvement relative to other baselines.

The estimates are $A_{\mathrm{E}} = 0.903 \pm 0.305$ and $A_{\mathrm{O}} = 0.937 \pm 0.529$, using maximum marginal working errors. The loading-zero conditional test leaves the solid-Earth scale free and gives $P = 0.1875$; the joint unit-point test gives $P = 0.8125$. The latter does not establish separate loading detection. The same six-period assumptions and selection limitations described below apply. An independent SVD implementation checks coefficients, covariances and these tests; this is numerical validation, not independent physical calibration.

The historical summer response used within-segment demeaning, 1,200-s triangular inference windows and precision-weighted combination of segment amplitudes.[3] The archived public-model chain already included a dominant degree-2 solid-Earth tide with real Love numbers and an eight-constituent, displacement-based loading approximation; separate solid-only diagnostics also exist. The present analysis resolves physical frequency groups and tests cross-period amplitude transfer using a different loading construction, while maintaining the provenance of each result. The archived configuration is not assigned wholesale to every earlier developmental calculation.

Joint estimation and uncertainty

All fits use equal weights for the retained 600-s averages, consistent with this reduction route. Columns are normalized for rank checks and QR solution; reported coefficients use their original units. SVD supplies an independently implemented numerical check. Rank loss or numerical near-degeneracy is reported rather than resolved with an arbitrary physical amplitude: the smallest singular value of the column-normalized design must exceed 10^{-11} times the largest. Zero nuisance columns after deletion are removed; loss of a target direction is an error.

For $y = L\theta + Z\gamma + \epsilon$, partial regression is implemented with both $M_Z y$ and $M_Z L$, where $M_Z = I - ZZ^{+}$. This is algebraically equivalent to the full joint fit.[26] Its noise covariance is $M_Z \Sigma M_Z^{\mathsf{T}}$; a residual is not a new independent observation. Least-squares covariance uses the full-model residual degrees of freedom. BJT-day and operating-period jackknives refit all retained parameters. For G deletion groups,

$$C_{\mathrm{JK}} = \frac{G-1}{G} \sum_{g} (\hat{\theta}_{-g} - \bar{\theta})(\hat{\theta}_{-g} - \bar{\theta})^{\mathsf{T}}.$$

We retain LS, delete-day and delete-period covariances separately.[27] The largest marginal standard error is a working summary, not a joint covariance or a calibrated confidence interval. The phase contours use delete-day covariance and $2F_{2,34}(0.95)$; their coverage is conditional and is not certified by the covariance ellipse alone.

Restricted wild diagnostics refit each candidate null, enumerate all $2^6 = 64$ operating-period sign patterns and recompute the delete-period studentizer for each realization.[28] All patterns are retained, including globally positive and negative ties. The two-sided minimum is $2/64 = 0.03125$. The procedure assumes approximately independent, sign-symmetric period errors; nuisance estimation prevents interpreting it as an exact experimental randomization test. It does not adjust for exploratory model selection. Scalar and joint-phase probabilities are distinct. No new confidence set is inferred from tests at zero and unity alone.

Independent scaled-SVD calculations check the twenty declared designs, including coefficients, covariances and both null tests. Joint and projected solutions agree numerically; eight fixed-calendar designs also reproduce the archived calculations. These are implementation checks, not physical calibration or arbitrary-noise coverage guarantees. The complete test ledger is retained with the code.

Validation and subtraction of a dominant component

A physical analysis may first examine the dominant constituent and then ask whether weaker predictions are supported. Agreement of a strong component with its expected amplitude and phase supports that component's model; it neither certifies all frequencies nor supplies an independent, error-free calibration. A fit to the whole template already contains the weak constituents and is not an independent strong-only subtraction. Constituent membership must be explicit, with each fixed remainder included once.

For a same-data decomposition $y = Sa + Wb + Zc + \epsilon$, let $s = M_Z S$, $w = M_Z W$ and $v = M_Z y$. The strong-first calculation uses

$$w_\perp = w - s(s^\mathsf{T} s)^{-1} s^\mathsf{T} w, \qquad v_\perp = v - s(s^\mathsf{T} s)^{-1} s^\mathsf{T} v,$$

$$\widehat{b} = (w_\perp^\mathsf{T} w_\perp)^{-1} w_\perp^\mathsf{T} v_\perp.$$

The strong and calibration coefficients are then recovered by back-substitution. This is block elimination of the joint model, not freezing the strong-only fit after a significance screen. The complete linear map from the original observations propagates covariance; every day/period deletion repeats all stages. If the strong component instead comes from independent calibration, its calibration uncertainty must enter the residual covariance; such an independent amplitude/phase calibration is not assumed here.[26]

We checked this construction by separating the member-matched M2 and N2 columns, keeping the near-12-h and diurnal scales, fixed remainder and calibration design unchanged. Their demeaned template RMS values are 3.254 and 0.686×10^{-18}. In the period-offset design, their correlation after removing the common nuisance space is 0.06880; projecting M2 then retains 99.527% of the already nuisance-

projected N2 energy. This fraction is not measured relative to the original, unprojected N2 template. Subtracting only the strong-only fit from the response and regressing on the unprojected weak column would attenuate the weak coefficient by $1 - \rho^2$ in this two-column setting.

Both elimination orders and joint QR agree with independently checked SVD operators; full covariance differences are below 2.3×10^{-12} in the calculation units. The M2/N2 split is exploratory, without multiplicity adjustment or replacement of the common lunar reference estimate.

The separate M2 phase targets are its in-phase and quadrature coefficients; N2 retains an independent fixed-phase scale. Near-12-h and diurnal scales remain free. If the inherited M2 template is $s = a\cos\theta + b\sin\theta$, its quadrature is $q = a\sin\theta - b\cos\theta$, with the same nodal basis and source-member averaging. This represents a 90-degree phase rotation, not orthogonality under the observed sampling. The corresponding identities are checked directly against the frozen columns.

Without daily terms, the two M2 coefficients are $(0.998, -0.068)$ with marginal working errors $(0.123, 0.177)$; the joint zero and unit-point probabilities are 0.03125 and 0.96875. With period-specific daily terms, they are $(0.821, -0.009)$ with errors $(0.198, 0.285)$ and probabilities 0.125 and 0.6875. These exploratory tests use the same observations and do not replace the grouped lunar reference or adjust for model selection. Their SVD check independently verifies coefficients and LS covariance; the deletion and sign tests use the validated shared library. High unit-point acceptance is not a probability that the physical prediction is true or a test of equivalence.

Solar-day tests, display and transfer

Daily sensitivity columns contain exact 24-h and 12-h sine/cosine pairs, globally or separately by operating period or occupied calendar block. They make no independent claim about the disturbance's mechanism or measured amplitude. S2 and a free 12-h sinusoid share a direction; binding S2 to K2 may leave a nonzero group direction without separately identifying S2. Near-degeneracy is evaluated through projected target information as well as the whole-design condition number.

For a separate retrospective transfer check, the grouping was declared before this calculation: January/April/May (1,263 bins) versus June–July/early August/late August (1,642 bins). Only the target amplitude is transferred. Test-set nuisance coefficients are re-estimated under both zero and learned amplitude, including other tidal groups in lunar fits. The score is $100(RSS_0 - RSS_A)/RSS_0$, on identical test observations and nuisance spaces. All eight comparisons are reported (Extended Data Table 3). They are conditional predictive scores, not P values or independent replications, and do not connect absolute period levels. Different designs have different null RSS; improvements are not used to select a model. Independent scaled-SVD calculations reproduce all eight amplitudes and scores. Period norm shares normalize full-record nuisance-projected template energy; test-set target norms instead use a local projection. Neither is a noise-calibrated measure of independent information.

Figure 3a chooses the 48-h window with the largest retained source-member count, irrespective of fit residuals. The amplitude is learned from other periods; one nuisance level is fitted using the held-out period. That level is subtracted for display only, with no separate segment recentering. Eighteen-bin display means never cross effective-interval boundaries or gaps exceeding 900 s. No plotted residual, phase contour or goodness-of-fit change is counted as an additional independent detection.

Held-out scores compare the same observations and locally fitted nuisance design with zero amplitude or an amplitude learned from other periods. The period-offset design gives a 3.525% pooled RSS reduction; period-daily freedom gives 0.539%, improving four of six periods. The shared training data make these conditional transfer checks, rather than six independent replications.

Conditional checks of the six-period test

The reported marginal error and the studentizer in the period-level test are different constructions. For the period-daily lunar scalar model, the working error is 0.20548, whereas its delete-period studentizing error is 0.11135. At unit response, the observed squared statistic is 4.61815; the largest of the 62 non-uniform sign-pattern statistics is 3.59197. Only the two globally uniform patterns reach the observed value, producing $P = 2/64$. This explains the enumeration without interpreting the working error as an inverted test interval. The restricted nuisance fit differs between zero and unit nulls.

We examined the estimator on its frozen sampling under three explicitly synthetic noise scenarios: independent Gaussian noise with standard deviation 20×10^{-18}; independent white noise with standard deviation 10×10^{-18} plus an Ornstein–Uhlenbeck process with standard deviation $\sqrt{300} \times 10^{-18}$ and 12-h correlation time; and variance-normalized Student-t_7 innovations with MA(1) coefficient -0.15 and marginal standard deviation 20×10^{-18}. MA memory restarts at effective-interval changes or gaps of at least 900 s. These parameters define test scenarios and are not claims about measured noise.

Each scenario uses 512 realizations, seed 2026110501, with true in-phase response one, quadrature zero and other nuisance coefficients zero. The last choice follows linear equivariance conditional on fixed designs and identifiable full and deletion fits, not an assertion that experimental offsets vanish. The fixed theoretical remainder has already been removed in this simulation coordinate, as in the estimator. The same simulated errors are used across the three declared designs. Every realization retains all 64 restricted sign patterns and recomputes the period studentizer. Vectorized and directly enumerated P values agreed for the first realization of each design/scenario. The OU process evolves across actual elapsed time, including inter-period gaps, rather than being reset to enforce exact period independence.

True-value acceptance at the 5% threshold ranges from 95.12% to 97.46% (Extended Data Table 7). Monte Carlo standard errors are approximately 0.70–0.95 percentage points. These finite scenarios do not indicate pronounced undercoverage, but they do not certify non-stationary experimental noise, unmodelled corrections, template uncertainty or selection-adjusted inference. Marginal max-error in-

tervals and joint-phase acceptance are distinct quantities. No observation, reference model or nuisance parameterization was changed following these tests. Additional simulations assess the whole-template scalar and two M2 phase designs separately, using 512 realizations per scenario and seed 2026110503. True-unit acceptance is 94.92%–96.88%, with Monte Carlo standard errors of 0.77–0.97 percentage points. These checks neither replace nor pool the grouped-lunar tests (Extended Data Table 7). Errors are shared within each three-design batch, so its nine table rows do not represent nine independent simulated experiments.

The actual delete-day F contours displayed in Fig. 2b were also tested, separately from the wild procedure. With seed 2026110504, 512 realizations per declared noise scenario and the unchanged threshold $2F_{2,34}(0.95)$, joint true-point coverage ranges from 94.53% to 99.41% (Extended Data Table 8). These finite scenarios test the displayed construction without calibrating experimental tail probabilities, unmodelled systematics or selection-adjusted coverage. The maximum marginal SE, a coordinate interval, a day-based joint contour and a period-studentized test remain distinct summaries.

Experimental controls and longer time scales

The B2 diagnostic reuses archived unique member matching without interpolation and requires complete support for every retained bin. The 10, 100 and 1,000 Hz gross-excursion thresholds were introduced after monitor inspection in the earlier analysis; they are not preregistered experimental criteria. All are retained here. The 100 Hz configuration contains 1,519 bins, 910,041 nominal members and 20 BJT days, only from P6–P8. Frozen indices, times, member counts and period identities match the current independently reconstructed observations. The matching algorithm and hardware synchronization were not independently rerun or certified in this check.

Both the baseline and B2-augmented models are fitted on that same support. With 100 Hz screening and no daily terms, the whole-template scale changes from 0.66817 to 0.67390; the lunar scale changes from 0.82353 to 0.84074. These subset estimates are not the six-period reference. Paired deletion errors use the difference of the two fit operators, retaining their dependence. Day and three-period deletion errors are separately reported (Extended Data Table 5), without assigning calibrated confidence to three clusters. Adding daily terms changes the two scales by 0.0402 and 0.0393. The intermediate threshold was inherited, not chosen to minimize these changes.

B2 enters as a freely scaled empirical regressor. Its fitted coefficient is an association parameter, not a calibrated round-trip-to-clock transfer function. Its separate template projection uses optical normalization 193399200000000 Hz without a one-way factor of two. An arbitrary nonzero rescaling of a freely fitted B2 column leaves its regression space unchanged; a small change in the clock scale therefore does not validate the normalization or exclude tidal-frequency content in the monitor. Increasing the threshold from 100 to 1,000 Hz adds only one bin yet substantially changes the monitor projection. All

thresholds are reported, without assigning a hardware cause to that bin or treating a smaller response shift as stronger exclusion. The monitor constrains only its measured channel and overlap, not clocks, unmonitored branches or earlier periods.

No long-term stability estimate obtained after fit-based offset adjustment is included. Short-period offset models cannot infer seasonal levels they have removed. Previously explored seasonal bounds use a separate level-preserving model and incomplete long-period loading; they are not new detections or limits from the present reference fit.

Data, code and assistance

The reproducibility package contains frozen observations, source-member identities, all declared model results, verification records and figure-generation code. Numerical reproduction starts from the derived table, with the preceding independent raw-counter readback documented separately. The analyses use the same experimental observations; computationally independent implementations are not independent experiments.

Author-approved access to raw data, derived observations and complete correction records remains to be specified before submission. The local review package is not a public archival deposit. Code availability likewise requires an author-approved archival/access statement.

OpenAI Codex assisted numerical implementation, code review, figure generation and drafting. Authors remain responsible for scientific verification, authorship, disclosures and access statements. Automated reviews are not author approval.

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Correspondence details are to be supplied. This author-review PDF includes the Article, Methods and Extended Data; no separate Supplementary Information is required for the present version.

Extended Data

Extended Data Table 1 | Retained support by operating period. Members are retained nominal one-second counter rows, including repeated time labels. Hours are member count divided by 3,600, not verified continuous gate uptime or statistical effective sample size. BJT-day counts use bin-mean days. P3–P8 preserve the historical eight-period identifiers; the two 2025 commissioning periods are excluded from this analysis.

Period	Calendar stage	Bins	Members	Nominal hours	Intervals	BJT days
P3	January	403	241,474	67.08	9	4
P4	April	440	261,782	72.72	11	5
P5	May	420	250,768	69.66	13	4
P6	June–July	599	358,311	99.53	8	9
P7	August a	592	355,036	98.62	5	8
P8	August b	451	269,702	74.92	3	5

Extended Data Table 2 | All declared reference, component and sensitivity estimates. Marginal errors are working standard errors; P values are conditional two-sided six-period diagnostics. Phase rows report the in-phase marginal coefficient but use a joint two-parameter null test. They must not be read as scalar tests. All models use the same 2,905 observations.

Design	Target	Estimate	LS SE	Day SE	Period SE	P(0)	P(unit)
period whole	Whole	0.913	0.087	0.101	0.105	0.03125	0.56250
period lunar	Lunar	0.973	0.118	0.121	0.095	0.03125	0.81250
period phase	Joint phase	0.972	0.118	0.121	0.099	0.03125	0.90625
group whole	Whole	0.885	0.090	0.108	0.109	0.03125	0.40625
group lunar	Lunar	0.945	0.119	0.123	0.095	0.03125	0.53125
segment whole	Whole	0.832	0.097	0.120	0.102	0.03125	0.12500
segment lunar	Lunar	0.952	0.127	0.137	0.098	0.03125	0.59375
period whole daily global	Whole	1.062	0.111	0.128	0.158	0.03125	0.65625
period lunar daily global	Lunar	1.023	0.119	0.123	0.111	0.03125	0.81250
period whole daily period	Whole	0.706	0.174	0.215	0.106	0.03125	0.06250
period lunar daily period	Lunar	0.761	0.190	0.205	0.111	0.03125	0.03125
period phase daily period	Joint phase	0.754	0.191	0.201	0.100	0.03125	0.12500
10d whole	Whole	0.884	0.088	0.100	0.097	0.03125	0.31250
10d whole daily block	Whole	0.790	0.209	0.277	0.129	0.03125	0.28125
10d lunar daily block	Lunar	0.891	0.234	0.282	0.188	0.06250	0.56250
10d phase daily block	Joint phase	0.866	0.243	0.277	0.181	0.25000	0.71875
5d whole	Whole	0.886	0.088	0.101	0.097	0.03125	0.31250
5d whole daily block	Whole	0.924	0.291	0.383	0.232	0.06250	0.78125

Design	Target	Estimate	LS SE	Day SE	Period SE	P(0)	P(unit)
5d lunar daily block	Lunar	0.893	0.308	0.430	0.307	0.09375	0.71875
5d phase daily block	Joint phase	0.892	0.308	0.437	0.305	0.31250	0.90625

Extended Data Table 3 | Bidirectional transfer of an amplitude between three-period groups. Early denotes P3–P5 (1,263 bins) and late P6–P8 (1,642 bins). Only the target amplitude is learned from the training group. Test-set nuisance coefficients are estimated on identical test observations under zero and learned amplitude. Daily denotes period-specific 24-h and 12-h terms. RSS values use the squared 10^{-18} working units; reductions use unrounded values, $100(RSS_0 - RSS_A)/RSS_0$. These retrospective scores are not significance tests.

Model	Train → test	Train A	RSS_0	RSS_A	Reduction (%)
Whole	Early → late	1.087	805576.4	796730.2	1.098
Whole	Late → early	0.704	475418.0	443953.3	6.618
Whole + daily	Early → late	0.539	794135.5	788831.8	0.668
Whole + daily	Late → early	0.782	430625.8	429618.6	0.234
Lunar	Early → late	1.051	803272.2	791400.7	1.478
Lunar	Late → early	0.872	451140.6	439756.4	2.523
Lunar + daily	Early → late	0.542	793651.3	788832.9	0.607
Lunar + daily	Late → early	0.844	430191.2	429590.7	0.140

The early group contains 754,024 retained nominal one-second members and the late group 983,049; these counts do not certify continuous gate uptime. Positive gains do not establish equal amplitudes between groups. Operating-period labels and retained support are listed in Extended Data Table 1.

Extended Data Table 4 | Strong/weak constituent cross-check on unchanged observations. M2 and N2 have separate constant scales; near-12-h and diurnal scales are estimated jointly, with the same fixed remainder as the grouped lunar model. Uncertainties are max(LS, delete-day, delete-period) working errors. P values are conditional, two-sided six-period sign diagnostics with minimum 0.03125, without multiplicity or model-selection adjustment. No strong coefficient is fixed after a P-value screen. N2 is not individually resolved in these designs.

Design	M2	N2	M2 P(0)	M2 P(1)	N2 P(0)	N2 P(1)
Period	0.997 ± 0.123	0.541 ± 0.648	0.03125	0.93750	0.34375	0.34375
Segment	0.942 ± 0.135	1.137 ± 0.830	0.03125	0.46875	0.15625	0.87500
Period+daily	0.820 ± 0.200	-0.043 ± 0.852	0.03125	0.28125	0.96875	0.12500

Extended Data Table 5 | B2 association checks on identical matched observations. W denotes whole template, L grouped lunar, and +d period-specific daily terms. B is the baseline scale. The response change is augmented minus baseline, with paired delete-day SE in parentheses; the next column gives its paired delete-period SE. The final column is the monitor's template projection under optical normalization with max(LS, delete-day, delete-period) SE. These are working errors, not calibrated intervals or physical transfer bounds. Support is P6–P8 only; all three archived gross-excursion screens are shown.

Hz	Model	Bins	B	Change (day SE)	Period SE	B2 projection (max SE)
10	W	1476	0.697	0.0074 (0.0103)	0.0114	0.074 (0.087)
10	W+d	1476	0.736	0.0175 (0.0182)	0.0166	0.173 (0.144)
10	L	1476	0.835	0.0083 (0.0061)	0.0044	0.082 (0.108)
10	L+d	1476	0.823	0.0214 (0.0196)	0.0119	0.210 (0.165)
100	W	1519	0.668	0.0057 (0.0072)	0.0090	0.075 (0.152)
100	W+d	1519	0.726	0.0402 (0.0326)	0.0313	0.516 (0.433)
100	L	1519	0.824	0.0172 (0.0094)	0.0096	0.223 (0.192)
100	L+d	1519	0.820	0.0393 (0.0239)	0.0095	0.502 (0.354)
1000	W	1520	0.667	0.0004 (0.0049)	0.0073	-1.169 (2.592)
1000	W+d	1520	0.724	0.0006 (0.0368)	0.0399	-1.227 (4.390)
1000	L	1520	0.821	0.0019 (0.0141)	0.0067	-3.501 (5.912)
1000	L+d	1520	0.817	0.0019 (0.0340)	0.0231	-3.728 (5.734)

Extended Data Table 6 | Interpretation of model terms.

Term	Definition and role	Information not supplied
Deterministic tide	Prediction from adopted astronomy and Earth/load response	Independent certification of every model component
Constant template scale	Common response to a specified time series	A separately measured Love number or evolving physical state
Period/group/segment offset	Jointly estimated constant calibration nuisance	Measured absolute connection or within-period constancy proof
Exact daily sinusoids	Empirical disturbance sensitivity class	A measured mechanism, amplitude prior or unique S2 attribution
Residual covariance	Statistical description of unexplained variation	Identification of its physical source
Phase quadrature	Constant component-response diagnostic	A fitted shift of observed times or an independent experiment

Extended Data Table 7 | Synthetic conditional validation on the frozen sampling. Each scenario has 512 realizations; error units are dimensionless template scale. D1 denotes the period-reference scalar lunar fit, D2 its period-daily alternative, and D3 the two-coefficient period-daily fit. D4 is the whole-template scalar, D5 the M2-only phase fit, and D6 M2 phase with period-daily terms. D1–D3 use seed 2026110501; D4–D6 use 2026110503. Errors are shared within, not across, these three-design batches. Noise codes are G, independent Gaussian; O, white plus OU; and T, Student-t plus MA. Max-SE means max(LS, delete-day, delete-period). Bias, SD and marginal-coverage columns concern the scalar or in-phase coefficient. Wild acceptance is scalar for D1, D2 and D4, and joint for D3, D5 and D6. MC SE is a binomial Monte Carlo standard error in percentage points; it is not experimental uncertainty. Noise scenarios were locally specified before these simulations, after the experimental analysis; they were not prospectively preregistered and do not certify the real noise process.

Design	Noise	Bias	SD	Mean max-SE	1.96 max-SE coverage	Wild acceptance	MC SE
D1	G	0.0031	0.1141	0.1331	96.68%	97.46%	0.70
D2	G	0.0112	0.1795	0.2328	97.07%	97.27%	0.72
D3	G	0.0111	0.1784	0.2354	97.27%	96.48%	0.81
D1	O	-0.0089	0.2400	0.3004	98.44%	96.29%	0.84
D2	O	0.0126	0.4051	0.6001	99.22%	95.70%	0.90
D3	O	0.0155	0.4068	0.6090	99.41%	95.70%	0.90
D1	T	-0.0043	0.1017	0.1212	97.66%	95.70%	0.90
D2	T	-0.0024	0.1595	0.2076	98.05%	96.48%	0.81
D3	T	-0.0025	0.1596	0.2093	98.24%	95.12%	0.95
D4	G	-0.0002	0.0882	0.0963	95.51%	96.09%	0.86
D5	G	-0.0028	0.1229	0.1433	96.48%	94.92%	0.97
D6	G	-0.0025	0.1923	0.2666	98.44%	96.09%	0.86
D4	O	-0.0066	0.2114	0.2429	95.12%	96.68%	0.79
D5	O	-0.0031	0.2536	0.3230	97.85%	96.68%	0.79
D6	O	-0.0038	0.4239	0.7350	99.80%	96.48%	0.81
D4	T	-0.0004	0.0703	0.0898	99.41%	96.68%	0.79
D5	T	0.0004	0.1036	0.1293	97.85%	96.09%	0.86
D6	T	0.0029	0.1650	0.2365	98.83%	96.88%	0.77

Extended Data Table 8 | Conditional coverage of the displayed Fig. 2b phase contours. Each entry is the joint true-point coverage in 512 synthetic realizations, followed by its Monte Carlo standard error in percentage points. These use the actual delete-day covariance and unchanged $2F_{2,34}(0.95)$ contour threshold, seed 2026110504. G denotes independent Gaussian noise, O white plus OU noise, and T standardized Student-t plus MA noise as defined in Methods. These are not period-wild acceptance rates or real-noise coverage guarantees.

Noise	Period reference (%)	Period daily (%)	Ten-day daily (%)
G	94.53 (1.00)	95.70 (0.90)	96.29 (0.84)
O	97.07 (0.75)	98.05 (0.61)	99.41 (0.34)
T	95.51 (0.92)	98.05 (0.61)	97.07 (0.75)

Extended Data Table 9 | Corrections used in the 2026 response coordinate. Values of a and b are in 10^{-18} and enter the ratio as $(1 + a)/(1 + b)$, after conversion to fractional units. A range gives applied group-specific values, not uncertainty. The Sr entries combine the corrected experimental table's density, Zeeman and available AC terms with the inherited -0.67×10^{-18} term. The common inherited $\Delta G = -3.116 \times 10^{-15}$ is applied separately. Missing terms are omitted from this numerical coordinate rather than asserted to vanish physically.

Period	Applied a	Applied b	Incomplete entries
P3	-93.4981	-9.9 to -9.6	Sr AC
P4	-181.1900	1.8 to 1.9	Sr AC
P5	-181.1000	0.0	Sr AC; Yb (0 is a placeholder)
P6	-172.0400	1.7 to 1.9	No missing Sr table entry; see qualification below
P7	7.9300	1.9 to 2.0	No missing Sr table entry; see qualification below
P8	8.0700	1.9 to 2.1	No missing Sr table entry; see qualification below

For P3, a includes the ledger's positive 37.2-mHz feedback entry converted using an assumed 698-nm optical node; no such entry is applied to P4–P8. The same entry applies to the separately analysed September and October 2025 records. P3–P4 Yb values follow the corrected table; P6–P8 values follow the archived clock-parameter record. June is assigned to the June–July correction period. Having no missing table entry does not certify a complete clock calibration. Independent confirmation of online/offline compensation, actual frequency reference and within-period time dependence remains unavailable. These limitations are distinct from the checked arithmetic and source membership.